UNIVERSITY OF KELANIYA – SRI LANKA
FACULTY OF SCIENCE

Bachelor of Science – Management & Information Technology (Honours) Degree Examination – December / January 2024

Academic Year 2022/2023 – Semester II

Information Technology
INTE 22212/ INTE 22212(R) – Software Design Patterns and Frameworks

No. of Questions:4 **No of Pages:3** **Time: 2 1/2 hrs.**

*Answer **ALL** questions. When allocating time for questions, take the marks allocated into consideration. State any assumption that you make.*

*This is an **OPEN BOOK** exam.*

QUESTION 1 (25 Marks)

In a coffee shop, customers can order a base coffee and add various optional ingredients (e.g., milk, sugar, chocolate). The price of the coffee depends on the base type and the selected additions. You need a flexible system to calculate the price of coffee dynamically without creating a complex hierarchy of subclasses for every possible combination of coffee and ingredients.

1.1 What is the best design pattern for the above scenario. **[5 Marks]**

1.2 Define a **Coffee** interface with a method **getCost()** to return the price and **getDescription**() to return the details of the coffee. **[5 Marks]**

1.3 Create a **BasicCoffee** class implementing the Coffee interface. **[5 Marks]**

1.4 If there are different profit for each coffee, how you are extending the above design. **[5 Marks]**

1.5 A logging system is needed in an application to track events, errors, or debugging information. The logger must ensure all logs are written to the same file or output stream. Creating multiple instances of a logger could result in duplicate or inconsistent logging output. Propose a design pattern for the logger system. Propose a design pattern. **[5 Marks]**

QUESTION 2 (25 Marks)

Case Definition: Implementing a Library System Using Micro services

The goal is to develop a scalable, maintainable, and efficient library management system using a micro services architecture. The system will handle core operations such as managing books, users, borrowing, returning, and notifications. Each micro service will address a specific domain within the library system, ensuring modularity and flexibility.

Following are the key functional requirement.

Book Management: Add, update, and search for books by title, author, genre, or ISBN.

User Management: Register new users, manage user accounts, and track borrowing history.

Borrowing and Returning: Facilitate book borrowing, renewals, and returns with due date management.

Notifications: Send notifications for due dates, new arrivals, and system updates.

Catalog Search: Enable users to search the catalog with advanced filters and real-time availability.

2.1 Draw a diagram to depict the above scenario. **[5 Marks]**

2.2 How can an organization effectively implement and manage a micro services architecture to achieve scalability, fault isolation, and ease of continuous deployment in the above scenario? **[5 Marks]**

2.3 Propose micro-service architecture for the above problem. **[5 Marks]**

2.4 What are the methods that you can propose to maintain the data consistency in the proposed micro service architecture? **[5 Marks]**

2.5 How would you achieve security in the above proposed micro-service architecture for the above problem? **[5 Marks]**

QUESTION 3 (25 Marks)

A document in a workflow system transitions through the following states:

- **Draft**: The document is created and can be edited. It can transition to the Moderation state.
- **Moderation**: The document is under review. It can either be approved (transition to Published) or rejected (back to Draft).
- **Published**: The document is finalized and cannot be edited. It can only transition to Archived.
- **Archived**: The document is stored permanently and cannot transition to any other state.

3.1 Define the state diagram for the above scenario. **[5 Marks]**

3.2 What is the best design pattern you can chose for the above scenario. Provide the reasons. **[5 Marks]**

3.3 Create a class diagram illustrating the use of the identified design pattern in the workflow system. **[10 Marks]**

3.4 Explain the use of Interface of OOP concepts for the above design. **[5 Marks]**

QUESTION 4 (25 Marks)

An e-commerce platform has a standard checkout process consisting of the following steps:

- **Validate Cart**: Ensure the cart has valid items and sufficient stock.
- **Process Payment**: Handle payment using different methods (e.g., credit card, PayPal, cash on delivery).
- **Arrange Delivery**: Set up shipping logistics, which vary based on the type of order (domestic or international).
- **Send Confirmation**: Notify the customer via email or SMS about the order status.

4.1 Draw a diagram to depict the above scenario. **[5 Marks]**

4.2 What is the best design pattern you can use for the above case? Explain your answer. **[5 Marks]**

4.3 It was decided to include exception-handling system. How would you handle the different types of exception handling? **[5 Marks]**

4.4 How Abstract classes can be used in the above scenario. **[5 Marks]**

4.5 How would you incorporate different levels of authorizations **[5 Marks]**

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